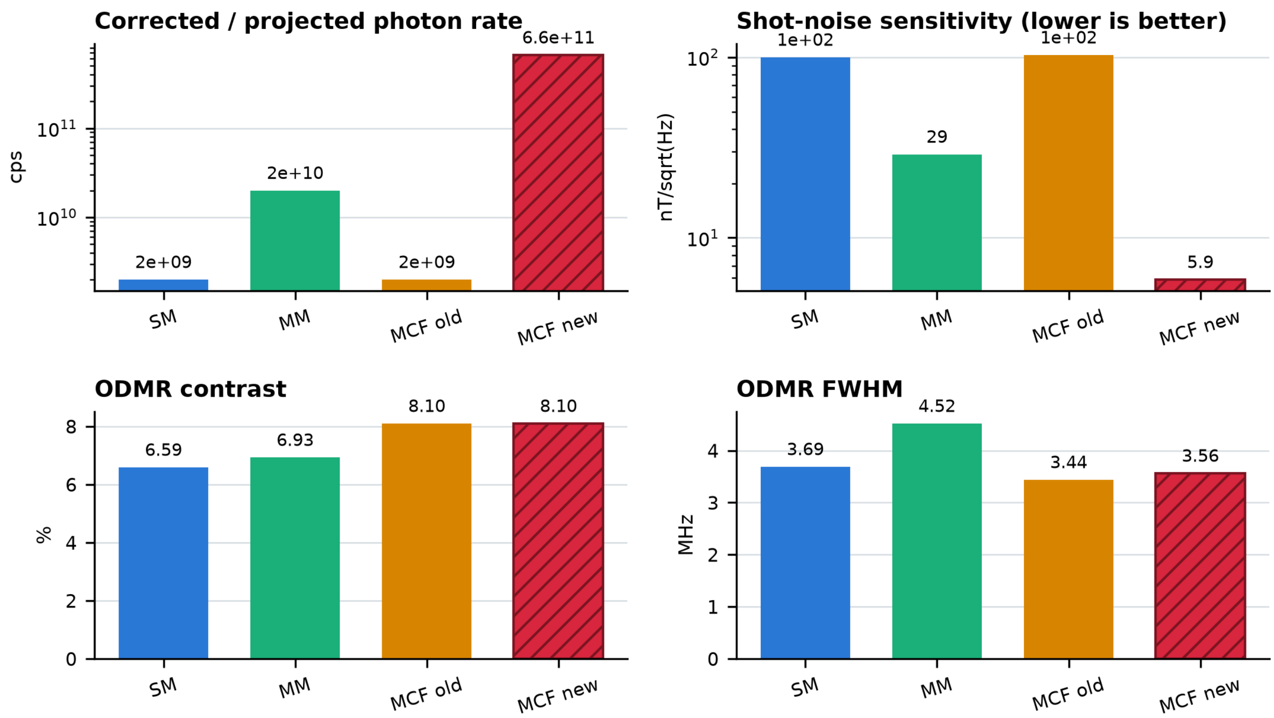
SM vs MM vs MCF optical probe comparison

Measured SM, MM and original MCF performance compared with the new monolithic seven-lens MCF model

Evidence boundary: SM, MM and MCF old are measured experimental values. MCF new is a shot-noise-limited optical-model prediction and has not yet been experimentally validated. Raw fiber photons and the empirical comparison normalization are reported separately.



Solid bars: measured experiment. Hatched bar: empirically normalized model target.

Primary comparison

Probe	Evidence	Photon rate*	Contrast	FWHM	Sensitivity
SM	Measured	2e+09 cps	6.59%	3.69 MHz	100.00 nT/sqrt(Hz)
MM	Measured	2e+10 cps	6.93%	4.52 MHz	29.00 nT/sqrt(Hz)
MCF old	Measured	2e+09 cps	8.10%	3.44 MHz	103.00 nT/sqrt(Hz)
MCF new	**Normalized model**	**6.63e+11 cps**	**8.10%**	**3.56 MHz**	**5.86 nT/sqrt(Hz)**

* Measured rates are corrected for ND attenuation. The new-MCF entry is a comparison-normalized count-rate proxy, not a physical detector conversion. Its separate raw output is reported below.

Main result

Under the explicitly labeled empirical normalization, the new design gives a 331.6x count-rate proxy and 17.6x lower shot-noise sensitivity than the measured original MCF, while retaining the measured 8.10% MCF contrast. The physical optical-model output is 6.05e+10 fiber photons/s, corresponding to 19.41 nT/sqrt(Hz) if one fiber photon is treated as one detected count. The measured MM remains the best demonstrated sensitivity at 29 nT/sqrt(Hz).

New MCF design and optical model

Printed structure	One monolithic IP-S body (n = 1.52): central lens plus six side lenses on a common support
Central channel	532 nm excitation; quadratic polynomial surface; 20 um aperture; 120 um protrusion
Side channels	650-850 nm collection; six biconic polynomial surfaces; 24 um aperture; 35 um core pitch; 10 um MFD
Best working gap	5.0 um central-lens-to-diamond clearance for the fixed optimized tip
NV ensemble	3 ppm NV centers throughout the 80-90 um layer in a 100 um diamond
Optical physics	3-D rays, local surface normals, vector Snell law, each Fresnel interface applied once, wavelength-dependent diamond index, saturation and excitation-collection overlap
Final quadrature	33 NV-depth planes and 33 trapezoid-weighted wavelengths

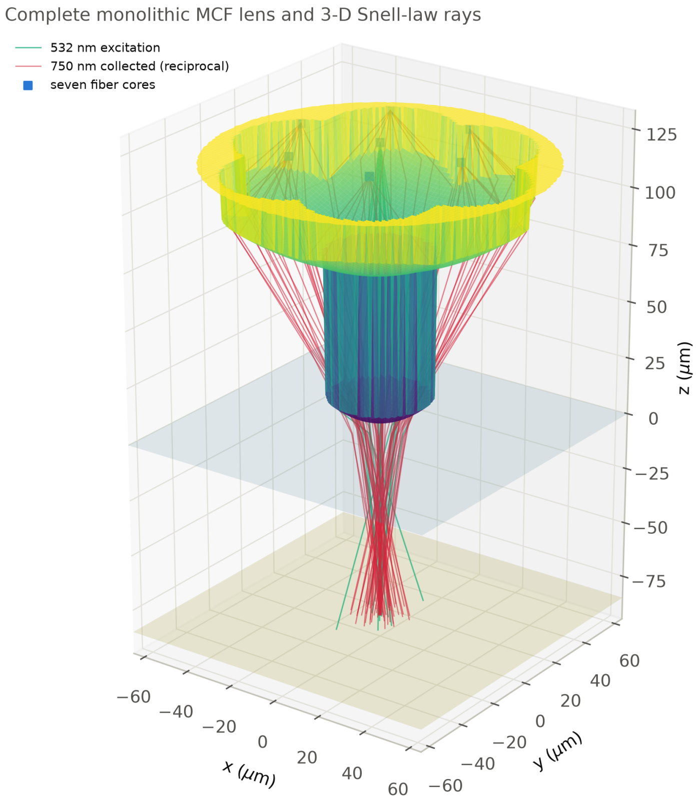


Figure 2. Complete seven-core printed-lens union. Green paths are 532 nm excitation rays. Red paths are reciprocal 750 nm collection rays. Refraction occurs only at the polymer-air and air-diamond boundaries.

Sensitivity model

Both outputs use eta_new = 103 x (FWHM_new / 3.44) x (0.081 / C_new) x sqrt(2e9 / R_new), with eta in nT/sqrt(Hz). For the raw optical result, R is 6.05e+10 fiber photons/s. For comparison with the measurements, R is the explicitly normalized proxy 6.63e+11 cps. The normalization factor is 10.97; because it exceeds one, it is not interpreted as detector efficiency.

Mounting tolerance and interpretation

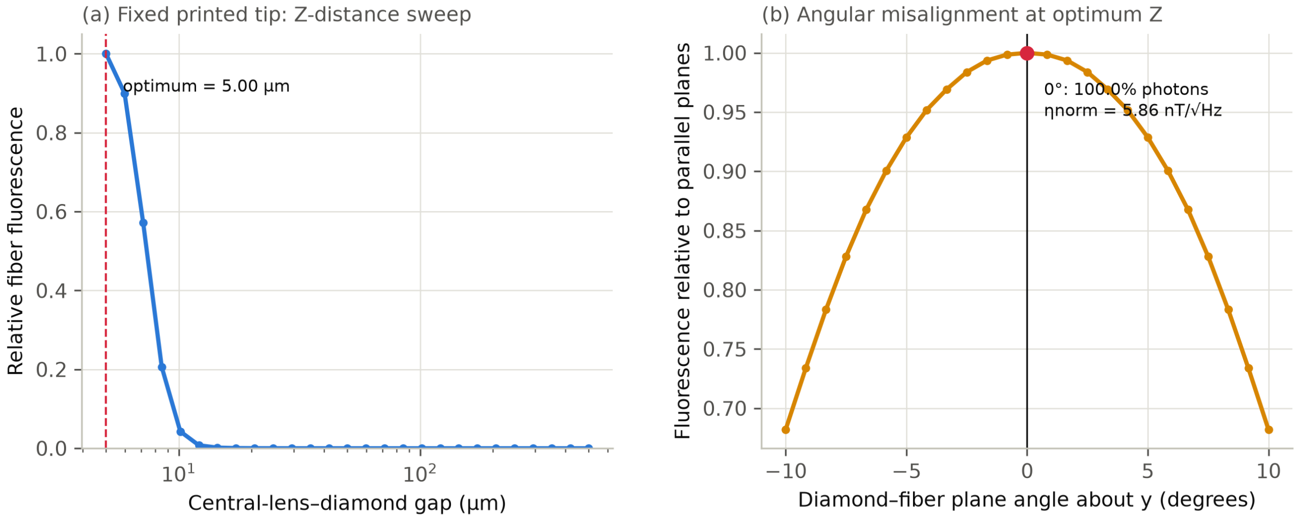


Figure 3. Fixed-lens Z-distance response and diamond-plane angular response. At nonzero tilt, all six side lenses are traced independently; the diamond and the 80-90 um NV layer rotate together about the fiber y-axis.

Z sensitivity	The sampled fixed-tip optimum is 5.00 um. At the sampled point nearest 10 um (10.15 um), modeled fiber fluorescence is 4.3% of the peak.
Angular sensitivity	At the sampled angles 5.00 and 10.00 degrees, predicted fiber-photon retention is 92.87% and 68.21%, respectively.
Measured ordering	SM and old MCF both give about 2e9 corrected cps. MM gives about 2e10 corrected cps and is therefore 10x higher in the supplied experiment.
Raw optical output	6.05e+10 fiber photons/s and 19.41 nT/sqrt(Hz) under the one-photon/one-count convention.
Normalized comparison	6.63e+11 cps proxy, 8.10% contrast, 3.564 MHz FWHM and 5.86 nT/sqrt(Hz).

Limits of the comparison

The new-design fiber-photon rate is an ideal smooth-surface geometrical-optics projection. The separate empirical normalization references the legacy old-MCF model checkpoint and must not be read as detector efficiency. The model does not include wave-optical diffraction at detailed freeform features, detector saturation or dead time, fabrication roughness and shrinkage, alignment drift, real wavelength-dependent core modes, fluorescence background, filter transmission or detector quantum efficiency. No background-derived contrast penalty is applied because no measured background-versus-footprint calibration was supplied.

Accordingly, compare the new MCF with the measured probes as a design target and expected trend, not as a demonstrated experimental result. The next decisive test is to fabricate the full STL, measure the unattenuated linear photon range, and repeat contrast/FWHM/sensitivity measurements at controlled gap and tilt.

Reproducibility

Inputs: figures/mcf_freeform_design.json and figures/mcf_gap_angle_sweep.json. Geometry: figures/mcf_freeform_full_seven_core.stl. Regenerate design figures with 'python redesign_fig.py --angle-deg ANGLE'. Generate this report with 'python make_comparison_pdf.py'.